

V. a. bidimensionale

$$f: D \rightarrow \mathbb{R}$$

$$\text{supp } f = \{x \in D \mid f(x) \neq 0\}$$

$$1_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

f dens. de prob. $f(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b] \\ 0, & \text{in rest} \end{cases} \Leftrightarrow f(x) = \frac{1}{b-a}, x \in [a, b] \Leftrightarrow f(x) = \frac{1}{b-a} \cdot 1_{[a, b]}^{(x)}$

It. $f(x) = \frac{1}{b-a} \cdot 1_{(-\infty, a]}^{(x)} \Rightarrow \dots F(x) = \begin{cases} \dots, & \text{if } x \in (-\infty, a] \\ 1, & \text{if } x > a \end{cases}$

T. lui Fubini

$$\iint_D f(x, y) dx dy, \quad D = [a, b] \times [c, d] \quad f(x, y) = g(x) \cdot h(y)$$

Atunci $\iint_{[a, b] \times [c, d]} f(x, y) dx dy = \int_a^b g(x) dx \cdot \int_c^d h(y) dy$

$$E(h(X, Y)) = \iint_{-\infty}^{\infty} h(x, y) \cdot f(x, y) dx dy$$

X v. a. cont.

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

ex. $f(x, y) = K \cdot e^{-(x+3y)}, 1_{(0, \infty)}^{(x, y)}$

$1_{(0, \infty)}^{(x, y)} = \begin{cases} 1, & x \in (0, \infty), y \in (0, \infty) \\ 0, & \text{in rest} \end{cases} = 1_{(0, \infty)}^{(x)} \cdot 1_{(0, \infty)}^{(y)}$

$f(x, y) = \underbrace{K \cdot e^{-x}}_{g(x)} \cdot \underbrace{e^{-3y} \cdot 1_{(0, \infty)}^{(y)}}_{h(y)}$

contraex $f(x, y) = K \cdot x^y \cdot 1_{[a, b] \times [c, d]}^{(x, y)}$

OBS: $\int_0^{\infty} \int_0^{\infty} K \cdot (e^{-x} + e^{-2y}) dx dy$

$$= K \int_0^{\infty} \int_0^{\infty} \underbrace{e^{-x}}_1 dx dy + K \int_0^{\infty} \int_0^{\infty} \underbrace{e^{-2y}}_{e^{-2y} \cdot 1} dx dy$$

Aplicatie

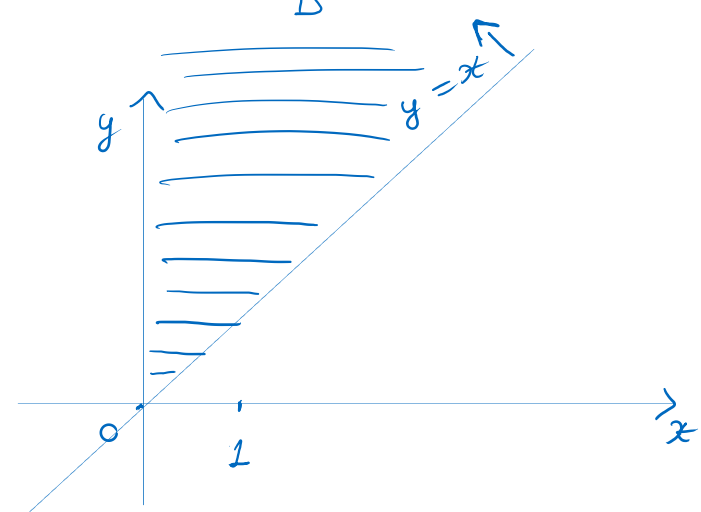
(X, Y) v. a. bidim. $f(x, y) = \begin{cases} 2 \cdot e^{-x} \cdot e^{-2y}, & x \in (0, \infty), y \in (0, \infty) \\ 0, & \text{in rest} \end{cases}$

a) $P(\underline{X > 1}, \underline{Y < 1}) = \int_1^{\infty} \int_0^1 2 \cdot e^{-x} \cdot e^{-2y} dy dx = 2 \cdot \int_1^{\infty} e^{-x} dx \cdot \int_0^1 e^{-2y} dy = 2 \cdot \int_1^{\infty} (-e^{-x})' dx \cdot \int_0^1 \left(-\frac{1}{2} \cdot e^{-2y}\right)' dy$

$$= 2 \cdot (-e^{-x}) \Big|_1^{\infty} \cdot \left(-\frac{1}{2}\right) \cdot e^{-2y} \Big|_0^1 = \left(0 - \frac{1}{e}\right) \cdot \left(\frac{1}{2} - \frac{1}{2}\right) = \frac{e^2 - 1}{e^3}$$

b) $P(X < Y) = \iint_{\Delta \{ (x, y) \mid x < y, x > 0, y > 0 \}} 2 \cdot e^{-x} \cdot e^{-2y} dx dy = \int_0^{\infty} \left(\int_0^y 2 \cdot e^{-x} \cdot e^{-2y} dx \right) dy = \int_0^{\infty} \left(e^{-2y} \cdot 2 \cdot (-e^{-x}) \Big|_0^y \right) dy = \int_0^{\infty} 2 e^{-2y} \cdot (-e^{-y} + 1) dy = \left[\int_0^{\infty} e^{-2y} dy - \int_0^{\infty} e^{-3y} dy \right]$

$$= 2 \cdot \left(-\frac{1}{2} e^{-2y} \Big|_0^{\infty} - \left(-\frac{1}{3}\right) \cdot e^{-3y} \Big|_0^{\infty} \right) = 2 \cdot \left(0 + \frac{1}{2} + \frac{1}{3} (0 - 1) \right) = 2 \cdot \frac{1}{6} = \frac{1}{3}$$



c) $P(X < a) = F_X(a)$

$$P(X < a) = \int_0^a \left(\int_0^{\infty} 2 \cdot e^{-x} \cdot e^{-2y} dy \right) dx = \int_0^a e^{-x} dx = 1 - e^{-a}$$